IT3120- Entrepreneurship for ICT Professionals

Process of Entrepreneurship in Sri Lanka, Opportunities and Challenges, and Success stories



Lecture Objectives

- Define entrepreneurship and entrepreneurial mindset
- Explain the entrepreneurship process
- Identify ICT entrepreneurship opportunities in Sri Lanka
- Analyze challenges faced by Sri Lankan ICT entrepreneurs
- Apply Lean Startup principles to technology startups
- Develop innovative ICT business ideas

Entrepreneurship

Entrepreneurship is the ability and willingness to identify opportunities, develop and organize a business, and manage its operations while taking on the associated risks and uncertainties in pursuit of profit.



Traditional Career Path vs Entrepreneurship

Traditional Career Path	Entrepreneurship
Works within an established organization	Creates and leads a new venture
Focuses on assigned responsibilities	Identifies opportunities and creates solutions
Earns a fixed salary and benefits	Has potential for greater financial rewards
Lower personal and financial risk	Higher risk but greater growth opportunities
Contributes to existing products and services	Develops new products, services, and business models
Career growth depends on organizational structure	Growth depends on innovation, leadership, and market success

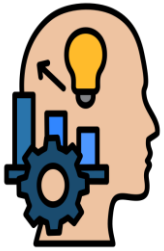
Entrepreneurship vs Intrapreneurship

Entrepreneurship	Intrapreneurship
• Entrepreneurship refers to the process of creating and managing a new business venture. Entrepreneurs identify market opportunities, develop innovative solutions, secure resources, and assume the risks associated with launching and growing a business. They have the freedom to make strategic decisions and shape the direction of the venture.	• Intrapreneurship refers to entrepreneurial activities carried out within an existing organization. Intrapreneurs act like entrepreneurs by generating innovative ideas, developing new products, improving processes, or exploring new business opportunities, but they do so using the organization's resources and within its strategic framework.

Entrepreneurship in IT

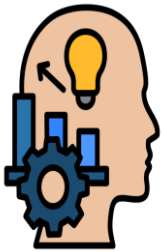
Entrepreneurship offers an alternative career path where ICT professionals become creators of technology-driven solutions, business founders, and innovators.





Key Characteristics of an Entrepreneurial Mindset

1. **Opportunity Recognition:** Entrepreneurs constantly observe their environment to identify unmet needs, market gaps, and emerging trends. They view problems as opportunities for innovation and value creation.
2. **Creativity and Innovation:** They think beyond conventional solutions and explore new approaches to solving problems. Innovation is often the foundation for creating competitive advantages and unique products or services.
3. **Calculated Risk-Taking:** Entrepreneurs understand that uncertainty is a natural part of business. Rather than avoiding risks, they assess potential outcomes and make informed decisions to pursue opportunities.



Key Characteristics of an Entrepreneurial Mindset

4. Resilience and Perseverance: Entrepreneurial journeys often involve setbacks, failures, and challenges. A strong entrepreneurial mindset enables individuals to learn from mistakes, adapt, and continue moving forward.

5. Customer-Centric Thinking: Successful entrepreneurs focus on understanding customer needs and delivering solutions that provide real value. They continuously seek feedback and improve their offerings based on customer experiences.

6. Continuous Learning and Adaptability: Entrepreneurs recognize that assumptions may not always be correct. They remain open to feedback, experiment with new ideas, and adjust their strategies based on what they learn from the market.

7. Vision and Leadership: An entrepreneurial mindset involves having a clear vision for the future and the ability to inspire others to work toward achieving that vision. Effective leadership helps transform ideas into successful ventures.

Entrepreneurship in the ICT Industry: Examples

Global Examples

Google: Started as an innovative search engine that provided more accurate search results. It later expanded into cloud computing, AI, mobile technologies, and digital services, becoming one of the world's leading technology companies.	**Facebook:** Began as a social networking platform connecting university students and grew into a global social media ecosystem, transforming how people communicate and share information online.
Uber: Developed a mobile platform that connects passengers with drivers, revolutionizing transportation services through technology and a platform-based business model.	**Airbnb:** Created an online marketplace that connects travelers with property owners, transforming the hospitality industry through digital innovation.

Entrepreneurship in the ICT Industry: Examples

Local Examples

99X: A leading Sri Lankan software engineering company that develops digital products for international clients, demonstrating global competitiveness in software development.

PickMe: A Sri Lankan ride-hailing platform that addressed local transportation challenges and later expanded into food delivery and logistics services.

Kapruka: One of Sri Lanka's pioneering e-commerce platforms, enabling online shopping and connecting overseas Sri Lankans with local markets.

Key Stages of the Entrepreneurship Process



Idea Generation

- The entrepreneurship process begins with the identification of a problem, need, or opportunity that can be addressed through an innovative solution. Entrepreneurs actively observe their environment, analyze customer frustrations, and explore emerging technological trends to generate business ideas.

Example: Identifying the difficulty farmers face in accessing real-time market prices and developing a mobile application to provide market information.

Key Stages of the Entrepreneurship Process

Opportunity Identification

- After generating an idea, entrepreneurs must determine whether it represents a viable business opportunity. This involves evaluating the market demand, target customers, competition, and potential value that the solution can provide.
- At this stage, entrepreneurs focus on validating whether the idea solves a meaningful problem rather than simply being a technically interesting concept.

Key Stages of the Entrepreneurship Process



Feasibility Analysis

- Feasibility analysis assesses whether the proposed business idea can be successfully implemented and sustained. This stage helps entrepreneurs minimize risks before investing significant time and resources.

Technical Feasibility	**Market Feasibility**
Operational Feasibility	**Financial Feasibility**

Key Stages of the Entrepreneurship Process



Business Planning

- Business planning transforms the idea into a structured roadmap for implementation and growth. A typical business plan includes:

  - Vision and mission
  - Problem statement
  - Proposed solution
  - Target customers
  - Value proposition
  - Revenue model
  - Marketing strategy
  - Operational plan
  - Financial projections
  - Risk assessment

Key Stages of the Entrepreneurship Process



Product Development and Validation

- Once the opportunity has been confirmed, entrepreneurs develop an initial version of the product or service. Rather than building a complete solution immediately, they often create a Minimum Viable Product (MVP) containing only the core features needed to test customer interest.
- Activities at this stage include:
  - Developing prototypes
  - Creating pilot versions
  - Testing with early adopters
  - Collecting customer feedback
  - Refining product features

Key Stages of the Entrepreneurship Process



Business Launch

- After validating the product, the entrepreneur officially introduces it to the market. The focus shifts from product development to customer acquisition and market penetration.
- Key activities include:
  - Marketing and promotion
  - Building customer relationships
  - Managing sales channels
  - Establishing operational processes
  - Monitoring customer satisfaction

Key Stages of the Entrepreneurship Process



Growth and Scaling

- Once the business achieves initial success, the focus moves toward expansion and long-term sustainability. Scaling involves increasing revenue and customer reach while maintaining operational efficiency.
- Growth strategies may include:
  - Entering new geographical markets
  - Adding new products or services
  - Expanding the customer base
  - Forming strategic partnerships
  - Attracting investors
  - Leveraging automation and digital technologies



In Class Activity 1

Multiple Choice Question

A startup team develops a basic mobile application for farmers and releases it to 50 farmers in one district to collect feedback before investing in additional features.
Which stage of the entrepreneurship process is primarily being performed?

A. Opportunity Identification
B. Business Planning
C. Product Development and Validation
D. Growth and Scaling



Multiple Choice Question

An entrepreneur identifies that many Sri Lankan tourists struggle to find reliable local tour guides. Before building a platform, she researches customer demand, competitor offerings, and willingness to pay.

Which stage is she currently performing?

A. Feasibility Analysis
B. Business Launch
C. Growth and Scaling
D. Resource Acquisition



Factors Shaping New Opportunities for ICT Entrepreneurs

Smartphone & Internet Access

Smartphones enable users to access mobile applications, online banking, e-commerce platforms, and digital services anytime and anywhere. Improved internet connectivity and expanding broadband and mobile network coverage have increased the number of people using online services.

AI Solutions (Artificial Intelligence)

AI Solutions involve developing intelligent systems that automate tasks, support decision-making, and improve efficiency. Opportunities include chatbots, recommendation systems, predictive analytics, image recognition, natural language processing, and business automation tools.

Factors Shaping New Opportunities for ICT Entrepreneurs

Digital Payments

The growing adoption of digital payment methods, such as mobile wallets, online banking, and QR-code payments, has made online transactions more convenient and secure. This supports the growth of e-commerce and digital business models.

Government Digitization

Government initiatives to digitize public services, records, and administrative processes have increased the demand for ICT solutions. Digital government services improve efficiency, transparency, and accessibility for citizens and businesses.



Global Market Opportunity

Unlike traditional businesses that often depend on physical locations, digital products and services can be delivered to users anywhere in the world through the internet, creating significant opportunities for growth and international expansion.

- **No Geographical Limits:** ICT products such as software, mobile applications, cloud services, and digital platforms can be accessed from anywhere with an internet connection
- **Scalable Globally:** Digital products can serve a large number of users with relatively little additional cost. Once developed, the same software or platform can be replicated and used by thousands or even millions of customers worldwide, enabling rapid business growth.
- **Lower Distribution Cost:** Unlike physical products, digital products do not require manufacturing, warehousing, or shipping. Software updates, applications, and online services can be distributed electronically, significantly reducing operational and distribution costs.

Kapruka.com Sri Lankan Opportunity Example

Opportunity Identified

- **Online Shopping Demand:** Kapruka recognized the growing demand for convenient online shopping services in Sri Lanka, enabling customers to purchase products and services without visiting physical stores.
- **Overseas Sri Lankan Market:** The company also identified an opportunity to serve Sri Lankans living abroad who wanted to send gifts, products, and services to family and friends in Sri Lanka.

Expansion

- **Gifts:** Kapruka expanded its offerings to include gift delivery services for various occasions, making it easier for customers to send gifts across the country.
- **Delivery:** The company developed efficient delivery and logistics capabilities to ensure products could be delivered conveniently and reliably to customers.
- **International Customers:** By leveraging digital technologies, Kapruka expanded its reach beyond Sri Lanka and served customers worldwide, particularly the Sri Lankan diaspora.

Key Lesson

Technology enables global reach. By identifying market opportunities and utilizing digital platforms, businesses can serve customers beyond geographical boundaries and create scalable growth opportunities.

Startup Challenges



Funding

- Securing adequate funding is one of the biggest challenges for startups. Entrepreneurs need capital for product development, technology infrastructure, marketing, operations, and talent acquisition.
- Many early-stage startups struggle to attract investors because they have limited market traction, revenue, or proven business models.
- As a result, entrepreneurs often rely on personal savings, family support, angel investors, grants, or venture capital funding.

Startup Challenges



Competition

- ICT startups frequently compete against established companies, global technology firms, and other startups offering similar solutions.
- Large competitors often possess greater financial resources, stronger brand recognition, and larger customer bases.
- To survive and grow, startups must differentiate themselves through innovation, superior customer experience, niche market focus, or unique value propositions.

Startup Challenges



Talent Retention

- Technology businesses depend heavily on skilled professionals such as software developers, data scientists, cybersecurity experts, and UI/UX designers.
- Startups often face difficulties attracting and retaining talented employees because larger organizations may offer higher salaries, better benefits, and greater job security.
- Building a positive work culture and providing growth opportunities are essential for retaining talent.

Startup Challenges



Market Uncertainty

- Entrepreneurs often launch products without complete certainty about customer demand or market acceptance.
- Customer preferences, industry trends, and economic conditions can change rapidly, affecting business performance.
- Startups must continuously gather customer feedback, test assumptions, and adapt their products and business models to changing market needs.

Startup Challenges



Technology Challenges

- Technology evolves rapidly, requiring startups to continuously update their products, skills, and infrastructure.
- must keep pace with emerging technologies, changing customer expectations, cybersecurity threats, and evolving industry standards.
- In addition, ensuring system reliability, scalability, data security, and integration with other technologies can be complex and resource-intensive, particularly for growing startups.

Startup Challenges



Regulatory Barriers

- ICT businesses must comply with various laws and regulations related to data privacy, cybersecurity, intellectual property, taxation, digital transactions, and industry-specific requirements.
- Understanding and complying with these regulations can be challenging, especially for startups with limited legal and administrative resources.

Lean Startup Solution

- The Lean Startup solution is a business approach created by Eric Ries for building a business by testing ideas with real customers before investing heavily in a finished product, replacing long business plans with fast, evidence-based learning through small experiments.

Lean Startup Solution



Risk Management for ICT Startups

Risk management is the process of identifying, assessing, and controlling potential events that could negatively affect a startup's objectives.

- **Technical Risks:** These risks arise from technology-related issues such as software defects, system failures, cybersecurity threats, scalability problems, or the use of unproven technologies. Technical risks can affect product quality, reliability, and customer satisfaction.
- **Financial Risks:** Financial risks involve insufficient funding, cash flow shortages, unexpected expenses, or lower-than-expected revenue. These risks can limit a startup's ability to operate, invest in growth, or sustain its business activities.

Risk Management for ICT Startups: Types of Risks

- **Operational Risks:** Operational risks are associated with internal business processes, people, and resources. Examples include skill shortages, employee turnover, project delays, infrastructure failures, and ineffective management practices.
- **Market Risks:** Market risks arise from uncertainty regarding customer demand, changing consumer preferences, competitive pressures, and economic conditions. A product may fail if it does not effectively address customer needs or gain market acceptance.
- **Legal Risks:** Legal risks involve non-compliance with laws, regulations, contracts, intellectual property rights, data protection requirements, and cybersecurity regulations. Failure to comply can result in penalties, lawsuits, or reputational damage.

Risk Mitigation Strategies



Pilot Testing

Before a full-scale launch, startups can introduce a pilot version or prototype to a small group of users. This helps identify technical issues, validate customer needs, and reduce the risk of investing heavily in an unproven solution.



Risk Register

A risk register is a document used to record identified risks, their likelihood, potential impact, and planned responses. It helps entrepreneurs systematically monitor risks and take proactive actions when necessary.



Backup Plans

Backup or contingency plans prepare the startup for unexpected situations such as system failures, funding shortages, cyberattacks, or operational disruptions. Having alternative solutions reduces the impact of unforeseen events and improves business resilience.



In Class Activity 2

Multiple Choice Question

A Sri Lankan startup has developed an innovative mobile application, but it cannot hire experienced software engineers because larger technology companies offer higher salaries and better benefits.

Which startup challenge is MOST evident in this scenario?

A. Market Uncertainty
B. Regulatory Barriers
C. Talent Retention
D. Technical Risks



Multiple Choice Question

A startup spends two years developing a sophisticated e-commerce platform. After launch, customer adoption is very low because users prefer existing platforms and do not see sufficient value in switching.

Which challenge was most likely underestimated?

A. Market Uncertainty
B. Operational Risk
C. Legal Risk
D. Financial Risk



Common Lessons from the Success Stories

- Identified real customer problems and market opportunities.
- Leveraged technology to create innovative solutions.
- Focused on customer value and service quality.
- Adapted and expanded based on market needs.
- Built scalable business models with long-term growth potential.



Summary

- **Entrepreneurship is the process of identifying opportunities, developing innovative solutions, taking calculated risks, and creating value through new business ventures.**
- **ICT professionals are uniquely positioned to become entrepreneurs** because they possess technical skills that can be transformed into scalable digital products and services for local and global markets.
- **The entrepreneurship process** involves Idea Generation, Opportunity Identification, Feasibility Analysis, Business Planning, Product Development & Validation, Business Launch, and Growth & Scaling.
- **Sri Lanka's growing digital economy** is driven by smartphone adoption, internet access, digital payments, government digitization, and a growing startup ecosystem, creating favorable conditions for ICT entrepreneurship.
- **Major opportunity areas for ICT entrepreneurs** include FinTech, EdTech, HealthTech, AgriTech, Tourism Tech, Artificial Intelligence, Cybersecurity, and E-Commerce.
- **ICT startups face several challenges**, including funding constraints, competition, talent retention, market uncertainty, technology challenges, and regulatory barriers, which require effective risk management strategies.
- **Successful entrepreneurs reduce risk through continuous learning** by testing assumptions, developing Minimum Viable Products (MVPs), gathering customer feedback, and adapting their solutions based on market needs.